

Jason Hernandez Lopez

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EXPERIENCE

Mechanical Build Engineer Intern

Albedo

May 2026 - August 2026, Broomfield, CO

- Authored the company's official 134-page Rev1 satellite bus build guide, defining the manufacturing sequence with build photos, CAD views, part/tool callouts, warnings, and verification steps.
- Designed, built, and tested custom satellite bus stands in 2 weeks using first-principles structural analysis.
- Developed production tooling for satellite bus assembly, including 4 indexers released as production parts and custom hardware used to retrofit four 1,100-lb motorcycle jacks.

Hardware Intern

NewHaptics

January 2026 - April 2026, Ann Arbor, MI

- Designed and built hardware for an automated imaging gantry system through structural design, lead-screw mechanism design, motor selection, machining, assembly, and custom test scripts.
- Built and tested electromechanical braille-module prototypes; designed hardware test setups to evaluate reliability, lifecycle performance, and prototype behavior.

Robotics Hardware Intern

Serve Robotics

May 2025 - August 2025, Miami, FL

- Scaled an operational robot fleet from 15 to 30 in 4 weeks and to 75 by week 10, averaging 10–12 electromechanical repairs/week across actuators, sensors, and PCBs using oscilloscopes, multimeters, and machine shop tools.
- Sustained $\geq 65\%$ fleet availability in the first 4 weeks and $\sim 80\%$ over the next 8 weeks by designing, building, and testing drivetrain reinforcements and mechanical fixtures.

Robot Technician

Starship Technologies

January 2025 - April 2025, Detroit, MI

- Served as the sole technician for 4 L4 autonomous robots, independently owning daily operation and maintenance.
- Performed electromechanical repairs and hardware upgrades across drivetrain, power, and sensor systems.

PROJECTS

- **STRYKER** – 6-DOF Robotic Arm: Mechanical Design · Electrical Design · Software Design
- **SENTINEL** – Self-Balancing Robot: Mechatronics · Control System Design · Engineering Analysis

EDUCATION

Bachelor of Science in Robotics Engineering

University of Michigan College of Engineering · Ann Arbor, MI · May 2027 · GPA: 3.6

SKILLS

Mechanical: SolidWorks, Creo, Fusion 360, GD&T, DFM/DFA, Mechanical Drawings, Machining, Rapid Prototyping.

Electrical/Test: Altium, EasyEDA, LTspice, PCB Assembly/Rework, Soldering/Crimping, Oscilloscope, Multimeter.

Programming/Controls: MATLAB/Simulink, Python, C++, Embedded Systems.